

USER MANUAL

GREEN SPACE MANAGEMENT SYSTEM

LAPR2



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1. Glossary

TEA (EN)	TEA (PT)	Description (EN)
Aerators	Arejadores	Agricultural implements used to enhance soil quality and promote healthy plant.
Agenda	Agenda	A calendar to appoint things.
Backhoes (loaders)	Retroescavadeiras	Machines used for excavating and loading materials.
Blowers	Sopradores	Devices that produce air at high pressure for cleaning or other purposes.
Bricklayer	Trolha/Pedreiro	Lay bricks, pre-cut stone and concrete blocks in mortar
Brush Cutters	Cortadores de relva	Tools used for cutting grass or low vegetation.
Budget	Orçamento	Estimate of costs for carrying out a project or activity.
Carries Out	Realiza	Execute or complete a task or activity.
Chainsaws	Motosserras	Tool used for cutting trees or wood.
Cisterns	Cisternas	Containers for storing water or other liquids.
Collaborator	Colaborador	Individual who works together with others to achieve a common goal.
Collaborators	Colaboradores	Individuals who work together to achieve a common goal.
Design	Design	Process of conceiving and planning objects, spaces, or systems.
Designer	Designer	Professional specialized in planning and creating spaces or objects according to aesthetic and functional criteria.

TEA (EN)	TEA (PT)	Description (EN)
Disc Harrows	Grades de Discos	An agricultural implement used primarily for soil preparation before planting crops.
Employees	Empregados	Individuals hired to perform a job in exchange for remuneration.
Entry	Entrada	An act of going or coming in.
Equipment	Equipamento	Tools, machines, or devices used to perform specific tasks.
Estimator	Estimador	Person responsible for calculating costs and resources required to carry out a project.
Garden	Jardim	Cultivated area with ornamental plants, flowers, trees, and lawns, usually in a residential setting.
Gardener	Jardineiro	Person specialized in caring for and cultivating plants in gardens.
Green Space	Espaço Verde	Area natural or urbanized intended for plant cultivation, leisure, or environmental conservation.
GSM (Green Space Manager)	Gestor de espaço verde	Person responsible for the administration and maintenance of green areas.
(Green Space User) GSU	Usuário de Espaço Verde	Person who uses or frequents green areas for leisure or other activities.
HRM (Human Resources Manager)	Gestor de Recursos Humanos	Professional responsible for coordinating activities related to the employees of an organization.

TEA (EN)	TEA (PT)	Description (EN)
IntelliJ/NetBeans	IntelliJ/NetBeans	Integrated development environment (IDE) used for software development.
Irrigation	Irrigação	Process of supplying water to plants in a controlled manner to keep them healthy.
Java	Java	Programming language first released by Sun Microsystems in 1995.
JavaDoc	JavaDoc	Java library used to generate useful documentation for java code.
JavaFX V	JavaFX II	Java library used to make graphics and media packages that enables developers to design, create, test, debug, and deploy rich client applications that operate consistently across diverse platforms
Lifting Platforms	Plataformas Elevatórias	Mechanical devices designed to elevate objects or individuals to higher positions.
Lighting System	Sistema de Iluminação	Set of devices used to illuminate a specific area.
Machine	Máquina	Mechanical or electrical device used to perform work.
Maintain	Manutenção	Act of preserving or repairing something to ensure its proper functioning.
Management	Gerenciamento	Process of dealing with or controlling things or people
Multifunctions	Multifunções	Equipment or devices capable of performing various functions.

TEA (EN)	TEA (PT)	Description (EN)
Object Orientation	Orientação a Objectos	Programming paradigm that focuses on modeling software as a collection of objects that interact with each other to perform tasks or represent concepts. The acronym for this concept is OO.
Object Serialization	Serialização de Objetos	Conversion of object's state to a byte stream so that the byte stream can be reverted into a copy of the object
Operating Machine	Máquina de Operação	Mechanical device used to perform specific tasks.
Park	Parque	Public area intended for leisure, recreation, sports, and contact with nature
Profission	Profissão	Occupation or activity for which a person receives payment.
Pruning	Poda	Act of cutting parts of a plant to control its growth and health.
Resources	Recursos	Means or materials available to perform a task or achieve an objective.
Rotating Machines	Máquina Rotativa	Equipment that performs circular movements around an axis.
Scarifiers	Escarificadores	Agricultural implements used for soil cultivation and preparation.
Semi-Urban	Semi-Urbano	Areas that exhibit both urban and rural characteristics.
Skills	Competências	Set of skills and knowledge required to perform a task or function.
Sprayers	Pulverizadores	Devices that spray liquids in a controlled manner.

TEA (EN)	TEA (PT)	Description (EN)
Task	Tarefa	Piece of work to be done or undertaken
To-Do List	Lista de tarefas	Record of activities to be completed within a specific timeframe.
Tractor	Trator	Motorized vehicle used to perform various agricultural or construction tasks.
Urban	Urbano	Related to urban areas, characterized by the presence of buildings and urban infrastructure
User Portal	Portal do Utilizador	Online platform where users can access information or perform specific actions.
VFM (Fleet Manager)	Gestor do Parque de Viaturas	Responsible for managing and maintaining the vehicle fleet.
Weeders	Capinadoras	Agricultural implements designed to control weeds in fields and orchards.

2. Introduction

This guide is designed to help any user on how to effectively use this product. With clear and simple guidelines, anyone will be able to understand and operate this product. The guide contains guidelines, pictures, and diagrams designed to help the user follow the steps and be able to do all the pretended tasks without needing help from the development team as the support of the application use.

The product described in this user manual is an application used by a green space company to manage their collaborators, machines, equipment and vehicles to their tasks with an easy and fast way to interact.

The application contains all the tools needed to manage, interact, or work with the company without difficulties or mistakes. It helps both the company teams and the managers of the company, or the ones who want to purchase a property.

This user manual has been provided to the user with the purpose of demonstrating how to use the application designed for the Green Space Company.

3. System Overview

The Green Spaces Management System is a comprehensive and efficient platform designed to facilitate the management and operation of green spaces. This software offers a wide range of functionalities to efficiently organize information about staff, vehicles, and other park resources.

The main features include:

- **Employee, Task, and Team Management:** The application allows for the registration and access to employees' personal data. Additionally, it enables viewing and managing tasks assigned to each employee according to their competencies. It also allows managing teams of employees, which are assembled according to the desired number of team members and the set of skills they possess.
- **Vehicle, Machinery, and Equipment Management:** This software enables the registration and monitoring of crucial vehicle information. Data such as maintenance history, records, and mileage are stored, ensuring the safety and proper maintenance of the park's fleet. Moreover, users can schedule and register maintenance check-ups for the vehicles and view a list of those scheduled for review. Additionally, this application allows for the management of various essential equipment required for park tasks.
- **Agenda Management:** The agenda is a crucial mechanism for planning the workweek. The system provides a view of tasks to be performed each day and allows for postponing or canceling tasks. By comparing agenda entries and the to-do list, it is possible to assess the work yet to be done, the week's workload, and the work completed by a team in a green space over a specific period and on a specific date.
- **Statistical Analysis:** The application allows analyzing water consumption, the most used equipment, and the profile of park users by age to manage public green spaces efficiently.
- **Irrigation System Management:** This system enables the planning and construction of irrigation systems to save materials such as piping, reduce the time needed for planning and installation, and create more efficient irrigation systems.
- **Emergency Planning:** This functionality allows green space managers to install signage for evacuating park users to a meeting point in emergency situations. The signage indicates the shortest route from various points in the park.

4. System Requirements

Compatible operating system

- macOS 10.15 or higher
- Windows 7/10/11
- Linux (Ubuntu 20.0.4 or higher)

Hardware Requirements

- Disk Space:
 - Minimum of 100 MB of free space
- RAM:
 - Minimum: 2 GB
 - Recommended: 4 GB
- Processor Speed:
 - Minimum 1.8 GHz

Software Requirements

- Java Development Kit (JDK) version 11 or higher
- JavaFx Software Development Kit (SDK) version 11.0.1 or higher

5. Troubleshooting

Launching the Application

The application is in “.jar” format. To launch the application, you must have Java JDK and JavaFX SDK installed on your machine, as mentioned before in System Requirements.

1 . Open Terminal (macOS/Linux) or Cmd (Windows)

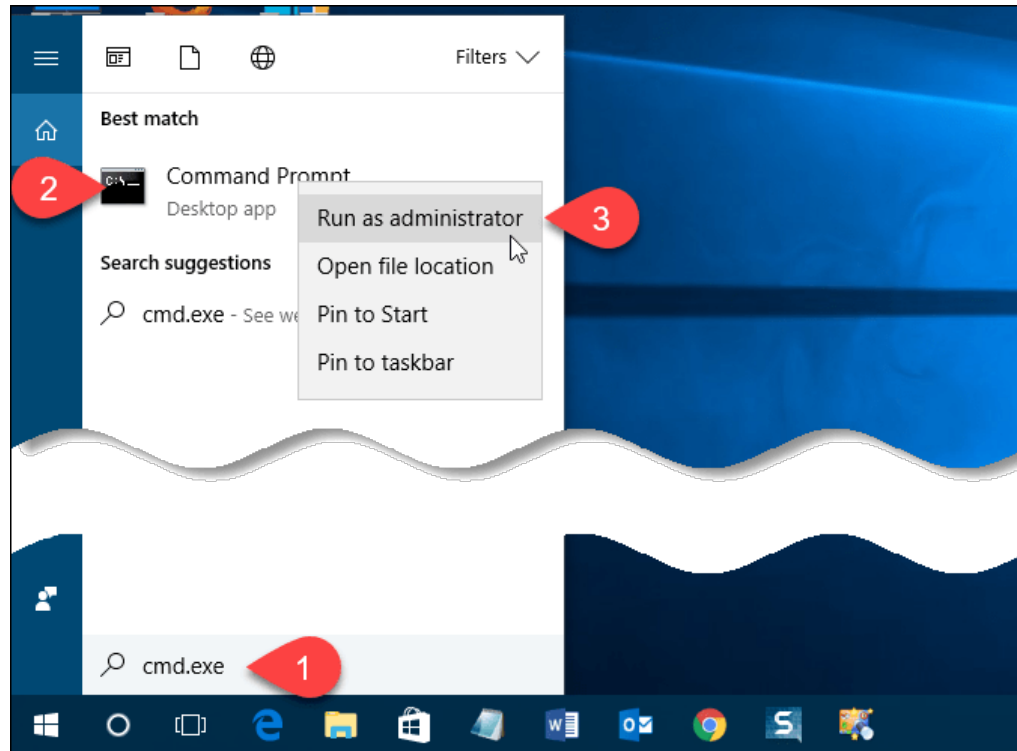


Image 1- Instructions for opening the terminal

2. Navigate to the program directory

3. Run the program by specifying the “.jar” file name

Setup for the first time

If you are trying to set up the system for the first time, log in as the user Main Administrator (Email: admin@this.app Password: admin). From there you can load a legacy system or register new employees to operate the system.

Authentication file not found

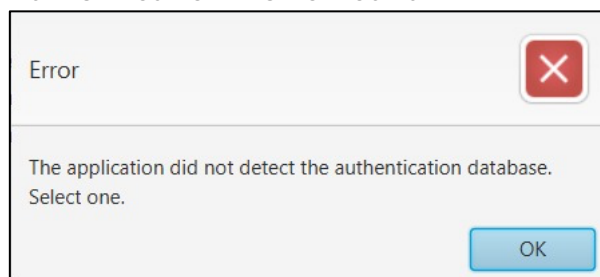


Image 2- Authentication file error

If you find one of these error messages during the initialization of the application, it's because the legacy files were not found in the same working directory of the program. To solve this issue, you can provide a new path to the files by clicking "PICK NEW PATH".

WARNING: If you just missing the authentication database file you will be able to run the program anyways, but you will not be able to do login to users created in previous sessions.

WARNING: If you do not provide a valid binary file your program will abort. To get access to this file please contact us.

WARNING: If you just missing the configuration file you will be able to run the program anyways, but you will not be able to input entries on agenda.

If you need a new configuration file, contact us to gain access.

Check for Resource Availability: Ensure that the resource exists and is correctly named. The file name is case-sensitive.

Configure Properties File

Inside the "target" folder, if you open the "config.properties" document, you will find the initial settings for the application as: the name of the organization, the time of work for collaborators, the sorting algorithm for the green spaces and the API to send emails for the assigned team to an entry. If you wish change the configure file, please contact us.

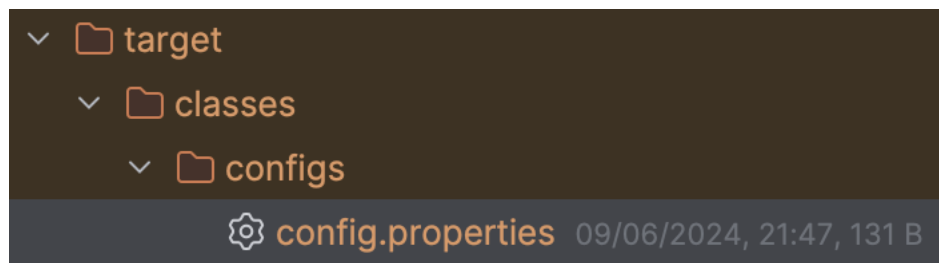


Image 3- Directory for the configuration file

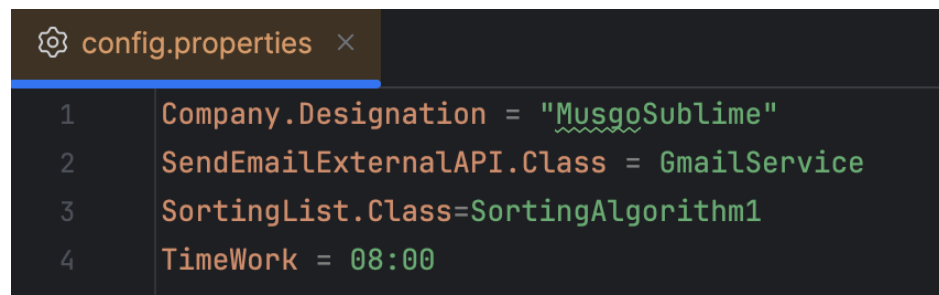


Image 4- Configuration File

Frequently Asked Questions (FAQ)

Are there any safety precautions I should be aware of?

The program does not have a good precaution for the safety of the Data. This is only a conceptual program.

Is it normal for the program to take so long to run?

The running of the program requires an elevated amount of processing to load all the data in the databases, it is normal to take a few seconds/minutes.

How can I contact customer support?

To customer support you need to contact the email: support@aquacode.com or contact the organization ISEP.

Can I use the Green Space Management app on my mobile?

No, the mobile app is not available.

Can I register another Main Administrator?

No, is not possible.

Can I register in the Green Space App without by registered on the System?

No, is not possible you need to be registred in the system first by the Main Administrator.

How many Green Spaces and Green Space Managers can have the System?

The system provides an unlimited space to register the number of green spaces that you want. And the same answer goes to the managers.

6. Features

6.1 Principal Login

When the Manager arrive at the Login Page of the Green Space Software

Step 1- Do login and enter the cards according syntax notes*:

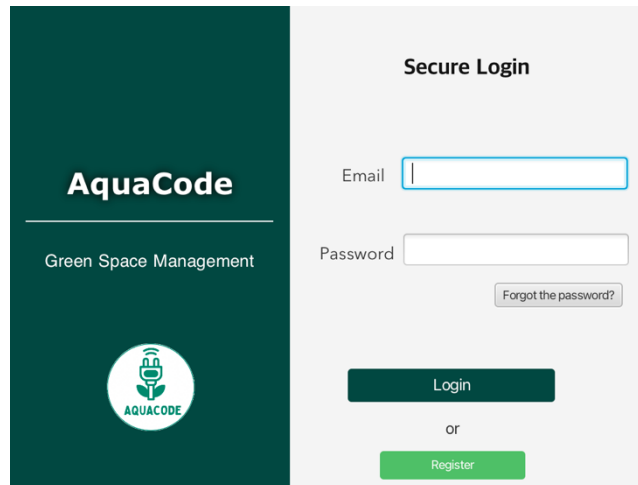


Image 5- Main menu "Secure Login"

Step 2- If you are not yet registered, click the "Register" button and this menu will appear where you must enter your credentials (email and password according syntax notes*):

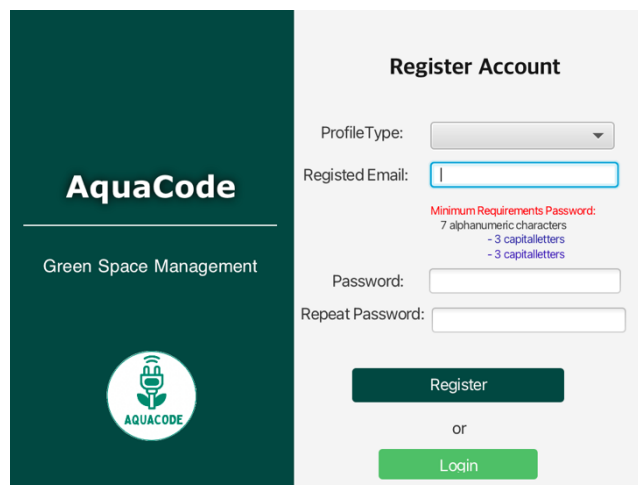


Image 6- "Register Account"

*Syntax Notes:

- Name: Should contain at least 8 characters.
- Email: Should contain an e-mail prefix, before the "@" symbol, and e-mail domain, after the "@" symbol.
- Password: Should contain at least 12 characters, with special characters included and upper-case characters.

6.2 Administrator Menu

When Administrator open and login in the software, (there are 8 possible paths), he can:

- Manage the Skills - The Manager can oversee and modify skills related to green space management.
- Manage Teams - The manager can create teams according to the number of members (maximum and minimum team size) and the set of skills that employees have.
- Manage the Jobs - The Manager can define and organize different job categories related to green space management.
- Manage Green Spaces- The Manager can regist a green space.
- Manage the Collaborators - The Manager can handle the administration and organization of collaborators involved in green space management.
- Manage the Vehicles - The Manager can oversee and administer vehicles used for green space management (including Check-Up List and Check Up register).
- Manage Tasks- The Mange can register tasks according to the title, description, duration, degree of urgency and the green space.
- Manage Agenda- according to the task to be performed, the start date and the start hour.

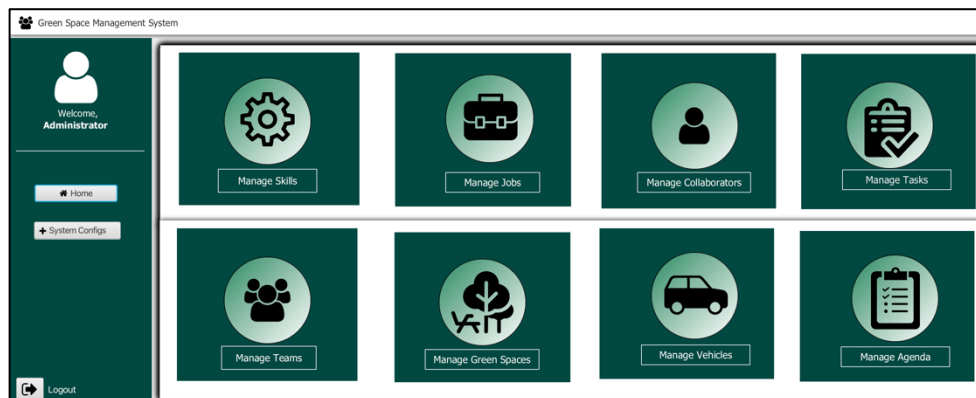


Image 7- Administrator “Home” menu

6.2.1 If the administrator clicks on the “Manage Skills” it will enter this menu where it can register skills according syntax notes*:

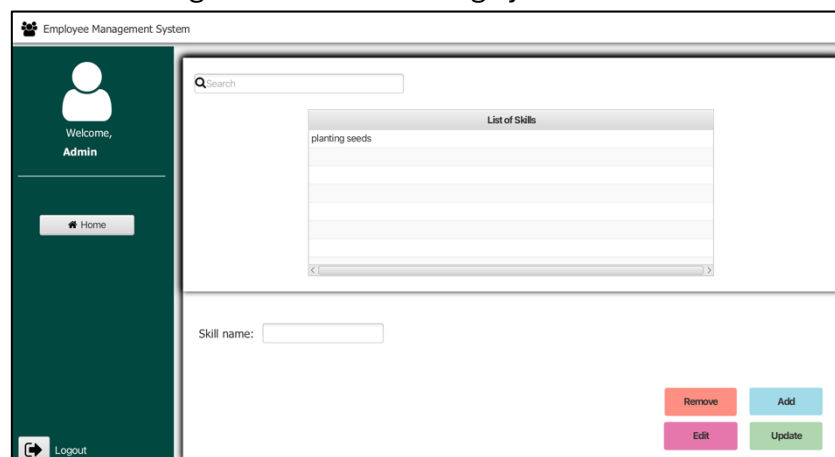


Image 8- “Manage Skills” Option

***Syntax Notes:**

- Skill name: A skill name can't have special characters or digits.
- Skill name: To register a skill is mandatory input the skill name.

When the administrator clicks on the “Remove” button he can remove a skill from the list. If he clicks the "Add" button he can add a skill to the list. If he clicks on the "Edit" button he can change the name of a skill that is already in the list and if he clicks on the "Update" button the menu shows the list with the added skills.

6.2.2 If the administrator presses the "Home" button again and clicks on the “Manage Teams” it will enter this menu where it can create teams according syntax notes*:

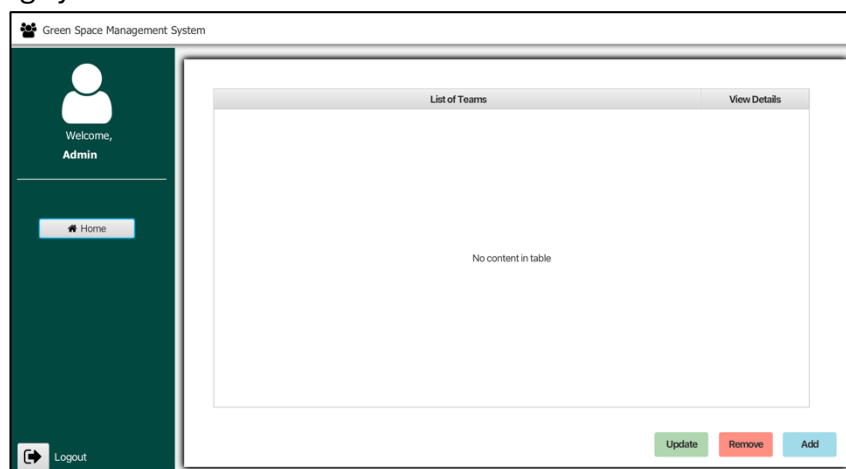


Image 9- “Manage Teams” Option

***Syntax notes:**

- Max Team Size: Contain the number maximum of collaborators that the team can have.
- Min Team Size: Contain the number maximum of collaborators that the team can have. The minimum size is at least 1.
- Skill Set: Should contain the skills that are needed to the team (not Mandatory)

When the administrator clicks on the “Add” button, this menu will appear where he can generate a team according to the collaborators’ skills.

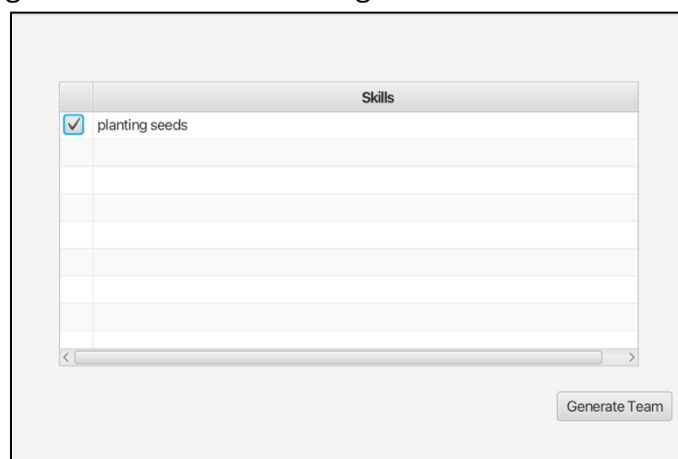


Image 10- Generate Teams according collaborators’ skills

If the administrator clicks on the "Generate Team" button, a new menu will appear where the admin can enter the team's name, size and number of collaborators per skill.

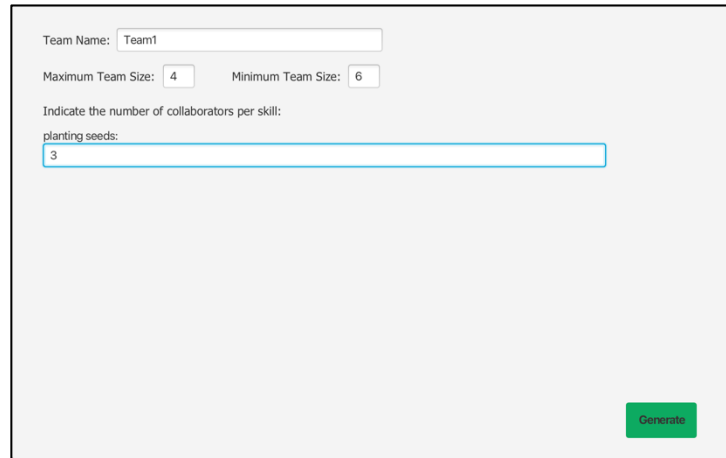


Image 11- Generate Teams according team name, team size and number of collaborators per skill.

Finally, if the administrator clicks on the "Generate" button, the team will be generated according to the desired information.

6.2.3 If the administrator presses the "Home" button again and clicks on the "Manage Jobs" it will enter this menu where it can register jobs according syntax notes*:

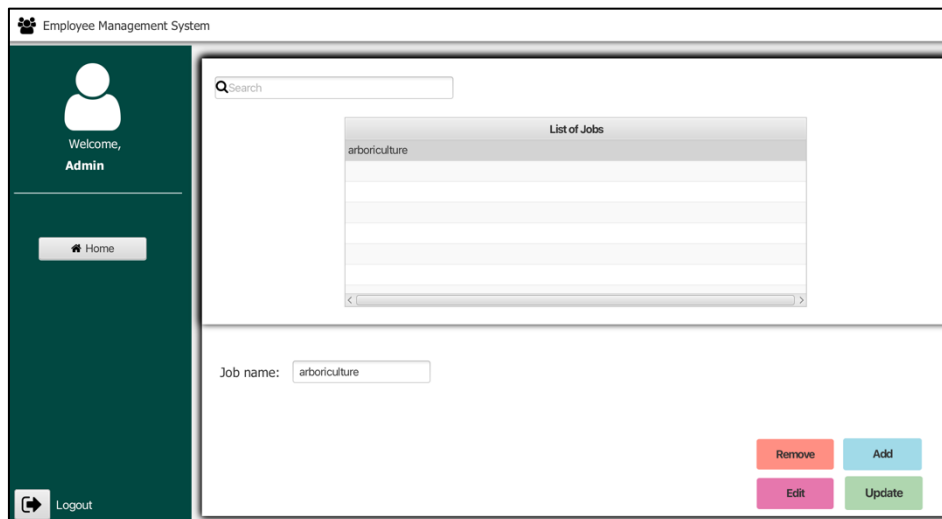


Image 12- "Manage Jobs"

***Syntax notes:**

Job name: To register a job is mandatory input the job name.

Job name: A job name can't have special characters or digits.

When the administrator clicks on the "Remove" button he can remove a job from the list. If he clicks the "Add" button he can add a job to the list. If he clicks on the "Edit" button he can change the name of a job that is already in the list and if he clicks on the "Update" button the menu shows the list with the added jobs.

When the administrator clicks on the “Remove” button he can remove a job from the list. If he clicks the "Add" button he can add a job to the list. If he clicks on the "Edit" button he can change the name of a job that is already in the list and if he clicks on the "Update" button the menu shows the list with the added jobs.

6.2.4 If the administrator presses the "Home" button again and clicks on the “Manage Green Spaces” it will enter this menu where it can register green spaces:

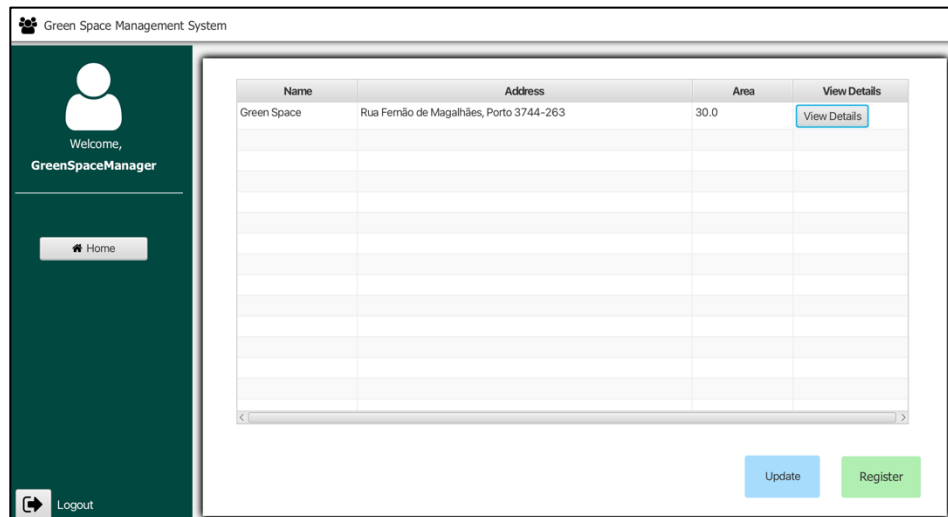


Image 13- “Manage Green Spaces”

When the administrator clicks on the “Register” button he can register a green space according to the green space name, address, zip-code, city, area and type.

Register Green Space

Name:

Address:

Zip Code:

City:

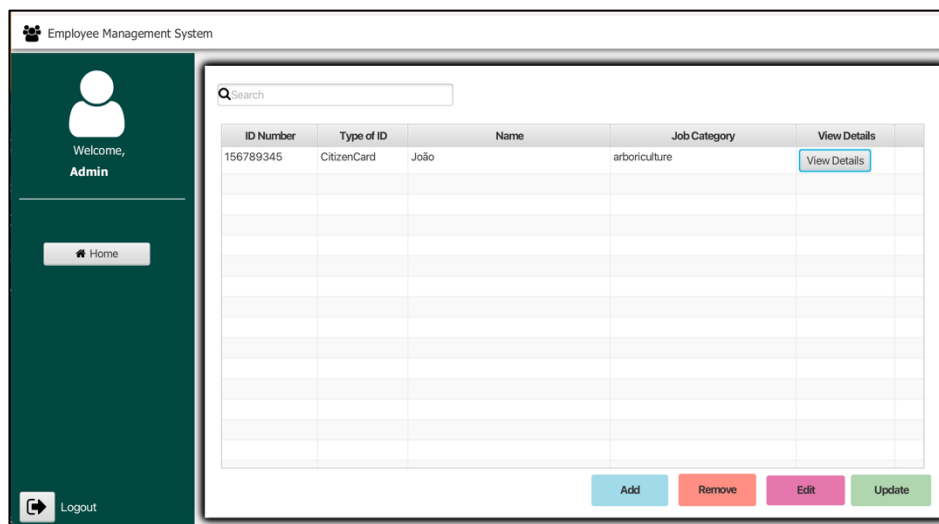
Area:

Type:

Register

Image 14- Register Green Spaces

6.2.5 If the administrator presses the "Home" button again and clicks on the “Manage Collaborators” it will enter this menu where it registers a collaborator according syntax notes*:



Syntax notes*:

- Name: Can have maximum of 6 words.
- Date of birthday: The age of the collaborator is mandatory be greater than 18.
- Admission date: Should contain at least 12 characters, with special characters included and upper-case characters.
- Address Street: (N/A)
- Zipcode: 4digits-3digits
- Address City: (N/A)
- Email: The email address needs to have a prefix, "@" and a domain for example: "mail.com" (the domain need to have one ".")
- Phone Number: Need to have 9 digits and have an international validation.
- ID doc type: Taxpayer number, Citizen Card or Passport
- Number of ID card: Need to be unique in the system and verified.
- Job Category: It is mandatory to select a job category.
- Skill(s): Not Mandatory.

If the administrator clicks on the "Remove" button he can remove a collaborator from the list. If he clicks on the "Edit" button he can change the collaborators information that is already in the list and if he clicks on the "Update" button the menu shows the list with the added collaborators.

When the administrator clicks on the “Add” button he can register a collaborator according to the collaborators’ name, email, date of birthday, admission date, address city, zip code, phone number, address street, job category, identification document, number of identification and skills for collaborators.

Image 16- Register Collaborators

6.2.6 If the administrator presses the "Home" button again and clicks on the "Manage Vehicles" it will enter this menu where it registers a vehicle according syntax notes*:

Image 17- "Manage vehicles" option

***Syntax notes:**

Format Plate Number: 00-00-XX (for vehicles between 1992 and 2005) , AA-00-AA (for vehicles after 2020), 00-AA-00 (for vehicles between 2005 and 2020)

Brand: It is the name brand of the vehicle, for example "Ford"

Model: It is the model of the vehicle, for example "F-150 Raptor"

Type: It is the type of vehicle, for example "Open Box"

Tare: It is the weight of an empty vehicle

Gros Weight: It is the total weight of a vehicle when is fully loaded with passengers, cargo or equipment.

Current KM: The current KM of the vehicle on that moment.

Register Date: When the vehicle was registered.

Acquisition Date: Date of the vehicle acquisition

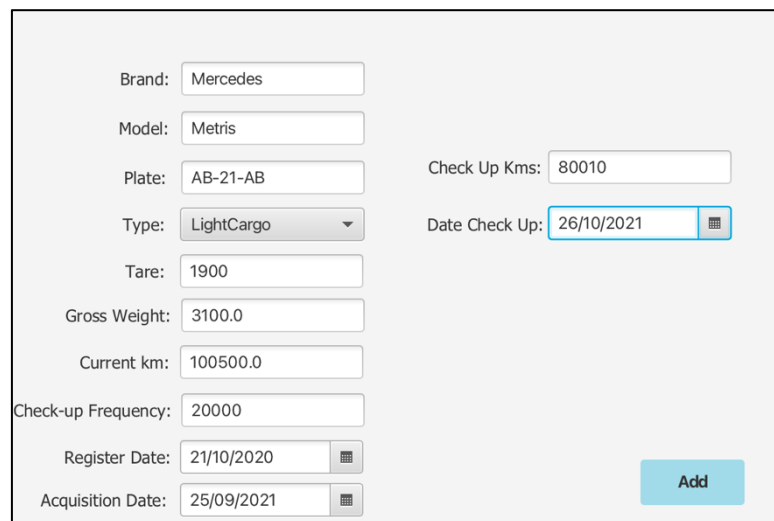
Maintenance/Check-Up Frequency: How many KM does the vehicle require maintenance.

Date of the Check-Up: Should be the current date or a past date, not a future date.

Current Kilometers: The odometer reading at the time of the check-up. This must be greater than or equal to the last recorded odometer reading to ensure accurate tracking of vehicle usage.

If the administrator clicks on the “Remove” button he can remove a vehicle from the list. If he clicks on the "Edit" button he can change the vehicle’s information that is already in the list.

When the administrator clicks on the “Add” button he can register a vehicle according to the vehicle’s brand, model, type, tare, gross weight, current kilometers, check-up frequency, register date and acquisition date:



Brand: Mercedes

Model: Metris

Plate: AB-21-AB

Type: LightCargo

Tare: 1900

Gross Weight: 3100.0

Current km: 100500.0

Check-up Frequency: 20000

Register Date: 21/10/2020

Acquisition Date: 25/09/2021

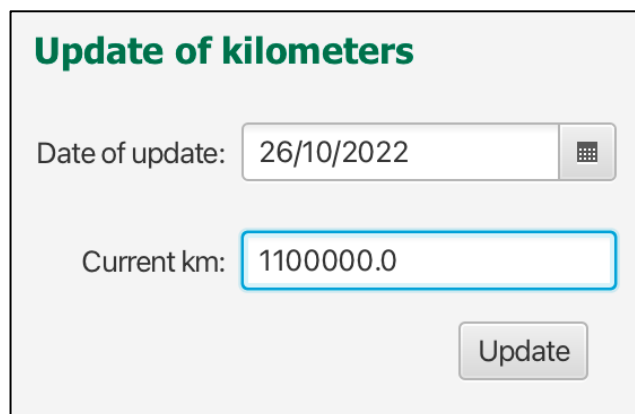
Check Up Kms: 80010

Date Check Up: 26/10/2021

Add

Image 18- Add vehicles

If the administrator clicks on the “Update kilometers” button he can update vehicle’s kilometers entering the date of update and the current kilometers:



Update of kilometers

Date of update: 26/10/2022

Current km: 1100000.0

Update

Image 19- Update Kilometers

If he clicks on the option “Vehicles Needing Check-Up Filter” the system will show the vehicles that need a check-up:

6.2.7 If the administrator presses the "Home" button again and clicks on the “Manage Tasks” it will enter this menu where it registers a task:

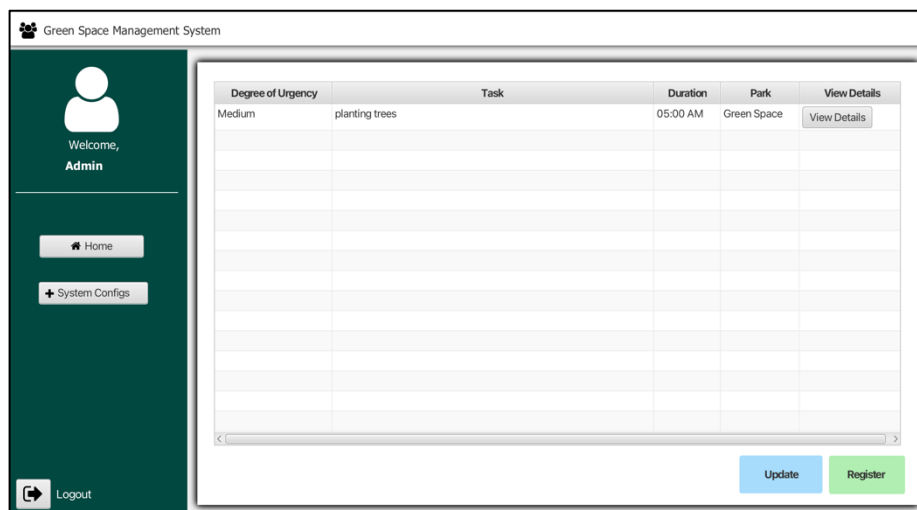


Image 20- "Manage Tasks" option

If the administrator clicks on the "Register" button he can register tasks according to the title, description, duration, degree of urgency and the green space.

Register a Task

Title:

Description:

Duration:

Degree of Urgency:

Green Space:

Register

Image 21- "Register a Task"

If the administrator clicks on the "Update" button the menu he shows the list with the added tasks.

6.2.8 If the administrator presses the "Home" button again and clicks on the "Manage Agenda" it will enter this menu where he can manage agenda entries:

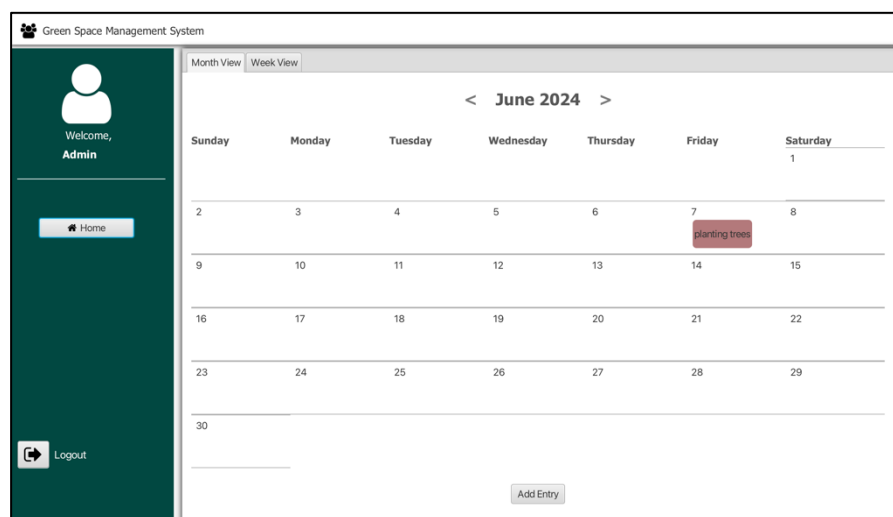


Image 22- "Manage Agenda" option

If the administrator clicks the “AddEntry” button it will enter this menu where he can add an entry in agenda list according to the task to be performed, the start date and the start hour.

Image 23- “Add Entry”

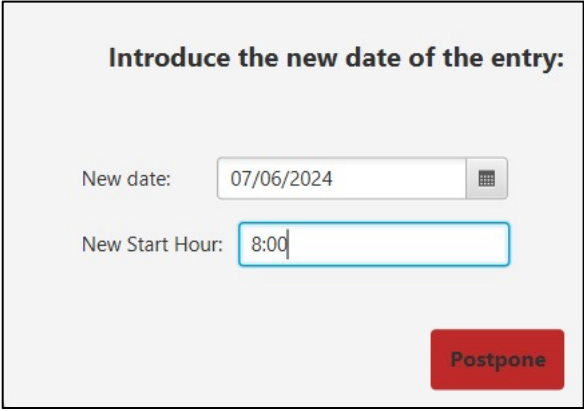
6.2.9 If the administrator exits the “Mana agenda” menu and re-enters, if he clicks on calendar, above the entry, this menu appears:

Image 24- Entry possibilities menu

6.2.10 If the administrator clicks on “Assign Team” button this menu will appear and this is where he can assign a team to the entry:

Image 25- “Assign Team” button

6.2.11 If the administrator come back to the Entry possibilities menu, if he clicks on the “Postpone” button this menu will appear, and this is where he can postpone the entry:



Introduce the new date of the entry:

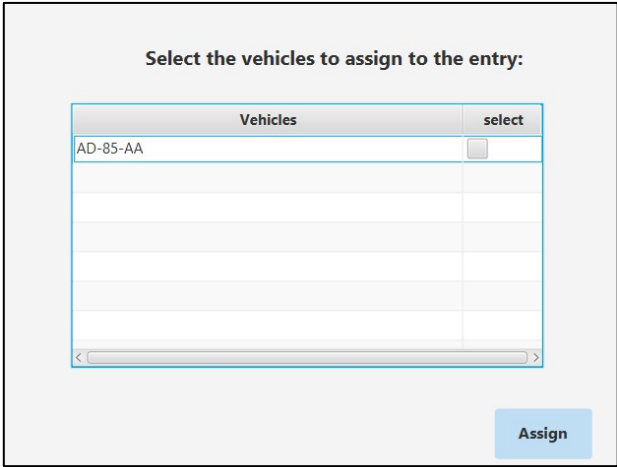
New date: 07/06/2024

New Start Hour: 8:00

Postpone

Image 26- “Postpone” button

6.2.12 If the administrator come back to the Entry possibilities menu, if he clicks on the “Assign Vehicle” button this menu will appear, and this is where he can assign a vehicle to the entry:



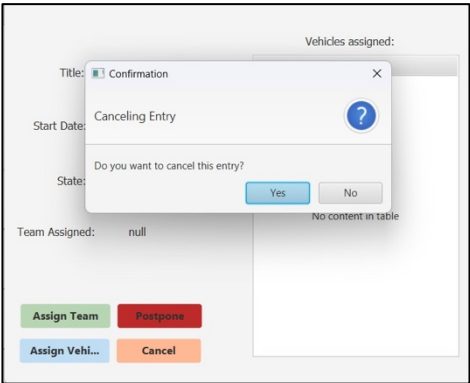
Select the vehicles to assign to the entry:

Vehicles	select
AD-85-AA	☐

Assign

Image 27- “Assign Vehicle” button

6.2.13 If the administrator come back to the Entry possibilities menu, if he clicks on the “Cancel” button he can cancel the entry and this message will appear:



Confirmation

Canceling Entry

Do you want to cancel this entry?

Yes **No**

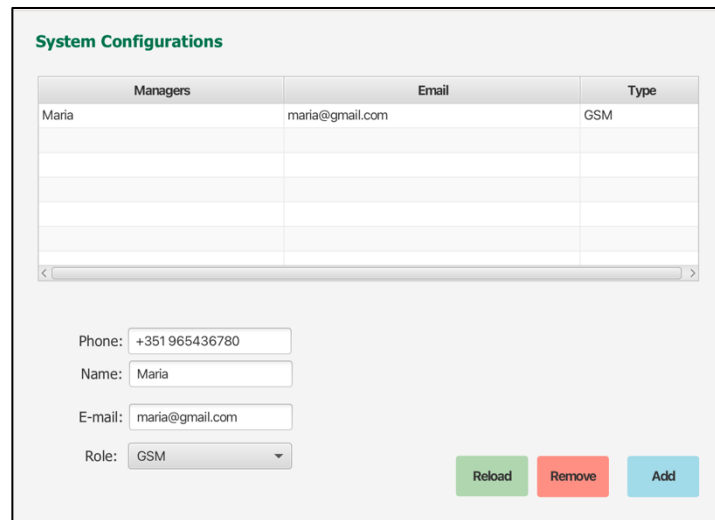
Vehicles assigned:

Start Date: State: Team Assigned: null

Assign Team **Postpone** **Assign Vehi...** **Cancel**

Image 28- “Cancel” button

6.2.14 In the main menu, if the administrator presses the “System Configs” button it will enter this menu where he can register a manager:



System Configurations

Managers	Email	Type
Maria	maria@gmail.com	GSM

Phone:

Name:

E-mail:

Role:

Image 29- “System Configs”

6.3 Human Resource Manager Menu

When GSM open and login in the software, (there are 4 possible paths), he can:

- Manage the Skills - The Manager can oversee and modify skills related to green space management.
- Manage Teams - The manager can create teams according to the number of members (maximum and minimum team size) and the set of skills that employees have.
- Manage the Jobs - The Manager can define and organize different job categories related to green space management.
- Manage the Collaborators - The Manager can handle the administration and organization of collaborators involved in green space management.

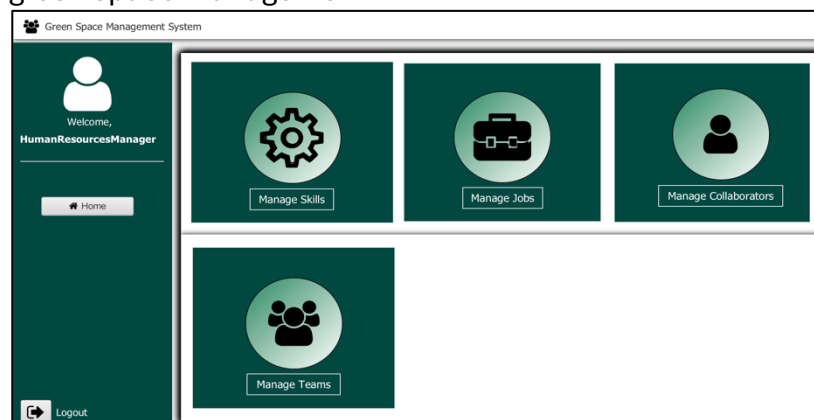


Image 30- HRM “Home” menu

IMPORTANT NOTE: When HRM clicks on the “Manage Skills”, “Manage Teams”, “Manage Jobs”, “Manage Collaborators” buttons, the functions of these buttons are the same as those of the Administrator when

pressing the same buttons in the Administrator "Home" menu, so just follow the instructions presented above

6.4 Vehicle Fleet Manager Menu

When VFM open and login in the software, (there are 1 possible paths), he can:

- Manage the Vehicles - The Manager can oversee and administer vehicles used for green space management (including Check-Up List and Check Up register).

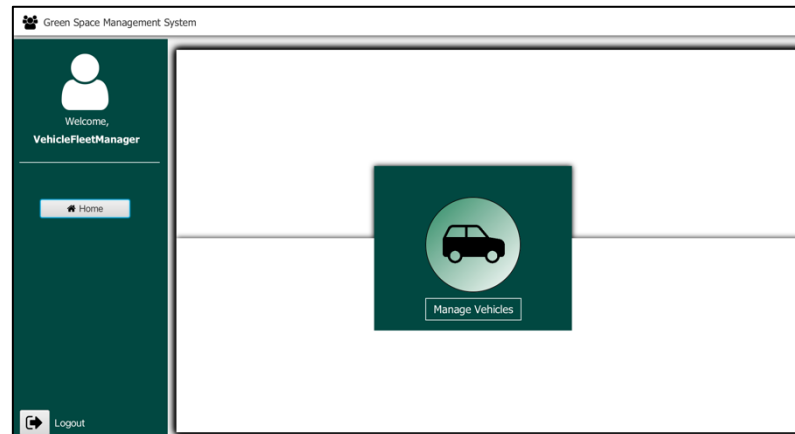


Image 31- VFM "Home" menu

IMPORTANT NOTE: When the VFM clicks on the "Manage Vehicles" button, the functions of this button are the same as those of the Administrator when pressing the same button in the Administrator's "Home" menu, so just follow the instructions presented above.

6.5 Green Space Manager Menu

When GSM open and login in the software, (there are 1 possible paths), he can:

- Manage the Skills - The Manager can oversee and modify skills related to green space management.
- Manage Teams - The manager can create teams according to the number of members (maximum and minimum team size) and the set of skills that employees have.
- Manage the Jobs - The Manager can define and organize different job categories related to green space management.
- Manage Green Spaces- The Manager can regist a green space.
- Manage the Collaborators - The Manager can handle the administration and organization of collaborators involved in green space management.
- Manage the Vehicles - The Manager can oversee and administer vehicles used for green space management (including Check-Up List and Check Up register).

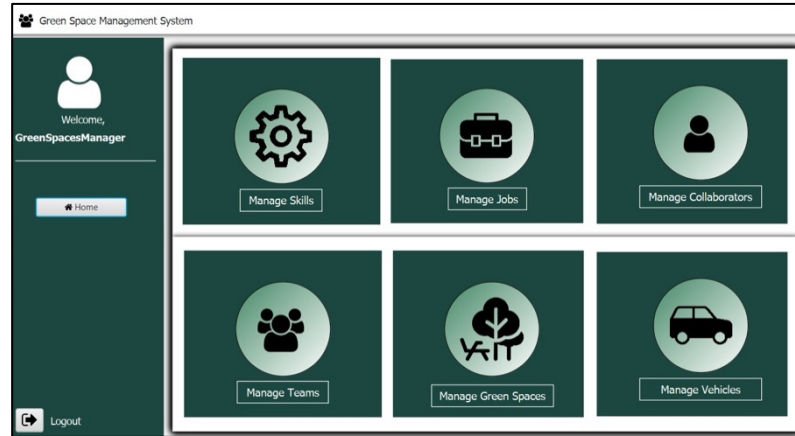


Image 32- GSM “Home” menu

IMPORTANT NOTE: When GSM clicks on the “Manage Skills”, “Manage Teams”, “Manage Jobs”, “Manage Green Spaces”, “Manage Collaborators” and “Manage Vehicles” buttons, the functions of these buttons are the same as those of the Administrator when pressing the same buttons in the Administrator "Home" menu, so just follow the instructions presented above.

6.6 Collaborator menu

When collaborator open and login in the software, he can select date to vie the tasks assigned:

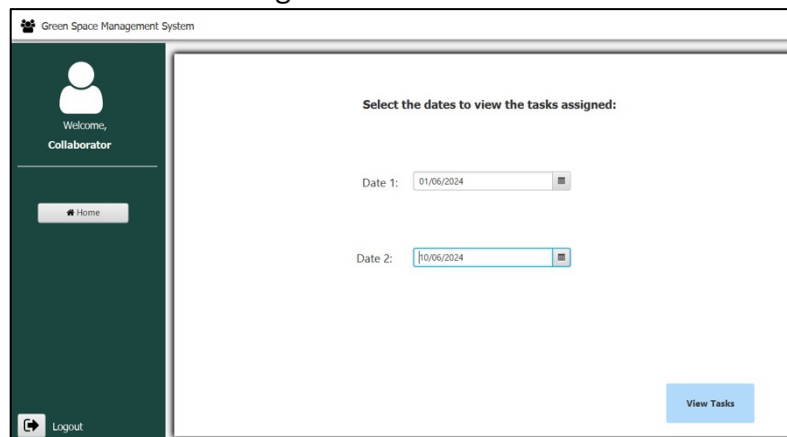
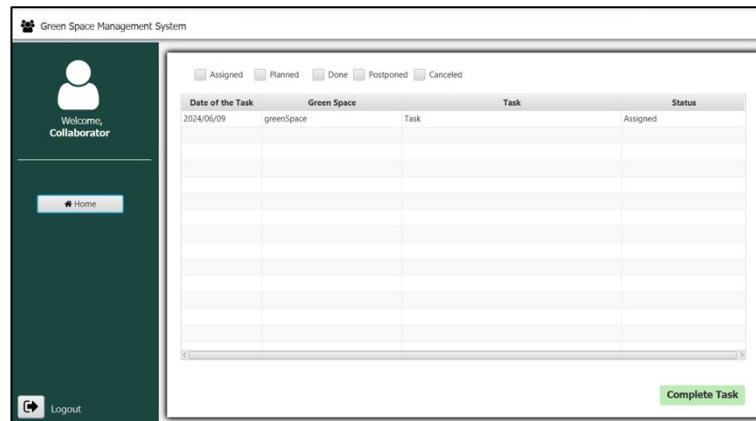


Image 33- Collaborator “Home” menu

If collaborator clicks on the “View Tasks” button he will see the entries assigned to him:



If collaborator clicks on the “Complete Task” button the status’ task switch to done.

7. Conclusion

Thank you for completing this user manual! By now, you should have a thorough understanding of our application. We hope this manual has equipped you with clear instructions, helpful insights, and practical tips to enhance your experience.

Throughout this manual, we've explored various facets of the application, including its features, setup process, operation, troubleshooting, and maintenance. Adhering to the guidelines in this manual will help you achieve optimal performance and extend the lifespan of the product or system. As a user, your role is crucial in maximizing the product's efficiency and effectiveness. Please refer to this manual whenever you have questions or need support. Additionally, our customer support team is available to assist you with any issues or questions not covered in this manual. We are committed to providing you with a seamless and enjoyable user experience. We value your feedback, so please share your thoughts on how we can improve. Your input is vital as we continuously strive to enhance our offerings and provide you with the best possible options.

Thank you for choosing the Green Space Management application.

Best regards,

Aquacode